

Chinmoy Biswas

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EDUCATION

National Institute of Scientific Research (INRS) 5 May 2025 - 2027 (expected)
Masters of Science (Quantum Machine Learning), CGPA: 4.3/4.3
Bangladesh University of Engineering and Technology (BUET) 7 Feb 2020 – 25 March 2025
B.Sc.Engg, (Electrical and Electronics Engineering), CGPA: 3.9/4.0 Major: Electronics

RESEARCH

(1) **Quantum Machine Learning – INRS (Master's Thesis)** 2025 - ongoing
Showing quantum advantage on generative AI tasks on superconducting and photonic commercial platform
(2) **Computational Photonic Modelling**
(A) **Photonic Memory using Phase Change Material – BUET (Undergraduate Thesis)** 2024 - 2025
Optical memory based forward and inverse design based on novel phenomena for neuromorphic photonics
(B) **Investigating Two-Photon Absorption Process in Si ridge waveguides – University of Alberta** 2025
Modelling non-linearity in silicon waveguides and Micro-Ring Resonators, in femtosecond laser irradiation
(C) **Thermal Emitter for applications in Solar Thermophotovoltaics – North South University** 2024 - 2025
Metamaterial Perfect Absorber-based Thermal Emitter for InSb TPV Cell
(3) **Modelling and Predicting from a Sparse Time Series Dataset** 2023 - 2025
Gaussian Interpolation and forecasting on a highly sparse load time series dataset (collected from BUET power plant)

EXTRA-CURRICULAR ACTIVITIES

(1) **Electrical & Software Project Manager for Team Interplanetar, BUET Mars Rover Team** 2021 - 2023
Designed the PCB of the Power Distribution Board for the rover. Managed a team of 12 for electrical component and inverse kinematic software development. Participated in the University Rover Challenge 2023 in **Utah, USA**, on a scholarship from **Grammenphone**. Instructed a course on Arduino and ROS.
(2) **IEEE BUET Student Branch** 2020 - 2024
Graphics Designer for **ICTP 2023** and **ICECE 2024** conferences held in BUET

WORK EXPERIENCE

(1) Industrial Trainee – **Neural Semiconductor** 2024

PUBLICATIONS

1. Biswas et al., *Opt. Exp.*, 33(20), 2025.
2. Biswas et al., *J. Appl. Phys.*, 139(9), 2025*.
3. Biswas et al., arXiv:2508.14069
*Editor's Pick

ACHIEVEMENTS AND AWARDS

(1) **VLSITHON Wins:**
● 2nd, ULKASEMI VLSITHON 2.0, RTL Section
● 3rd, ULKSEMI VLSITHON 2.0. Analog Section
(2) **BUET Scholarships:**
● RISE Undergraduate Research Grant
● Dean's List Award
● University Merit Scholarship
● Scholarship for work on Team Interplanetar
(3) **INRS Scholarships:**
● Master's Graduate Funding
● PBEEE Master's Scholarship
● FRQNT Master's Scholarship
● SPIE Master's Scholarship

PROJECTS

● SPICE Clone Using MATLAB
● ARM Single Cycle Processor Wall Art
● 12-bit ALU, RTL to GDS
● 8-Bit SAP1 Computer

SKILLS

Programming Languages: C, C++, Python, MATLAB, Shell, SystemVerilog **Software:** Cadence Virtuoso, Altium, PSpice, Quartus, uVision, Proteus, Simulink, ROS, Ansys Lumerical, COMSOL Multiphysics, CST Microwave Studio, AutoCAD **Frameworks:** Pytorch, Tensorflow, Pandas, NumPy, MEEP, STM32F446RE, Arduino

RELEVANT COURSES

● VLSI Circuits and Design
● Analog Integrated Circuits
● Digital Electronics
● Microprocessor and Embedded Systems
● Digital Signal Processing
● Power Electronics
● Semiconductor and Nano Devices
● Nanoelectronics and Nanotechnology
● Solid State Devices
● Theory of Computation